

Ex. No: 12 Page Replacement Algorithms

PROGRAM A: FIFO PAGE REPLACEMENT

C PROGRAM

```
#include <stdio.h>

int main()
{
    int pages[50], frames[10];

    int n, f, i, j, k = 0;

    int fault = 0, found;

    printf("Enter Number of Pages: ");

    scanf("%d", &n);

    printf("Enter Reference String:\n");

    for(i = 0; i < n; i++)

        scanf("%d", &pages[i]);

    printf("Enter Number of Frames: ");

    scanf("%d", &f);

    for(i = 0; i < f; i++)

        frames[i] = -1;

    for(i = 0; i < n; i++)

    {
        found = 0;

        for(j = 0; j < f; j++)

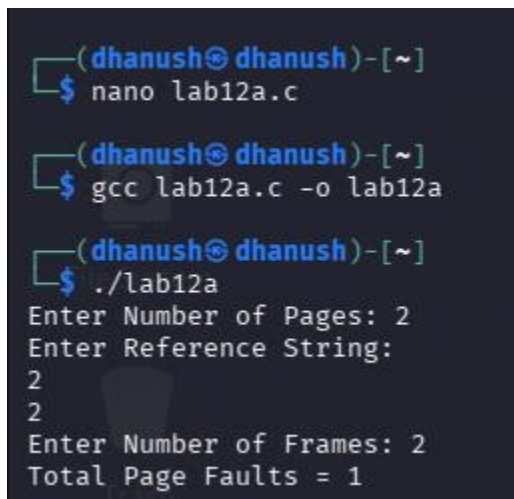
        {
            if(frames[j] == pages[i])
```

```

found = 1;
break;
}
}
if(found == 0)
{
frames[k] = pages[i];
k = (k + 1) % f;
fault++;
}
}
printf("Total Page Faults = %d\n", fault);
return 0; }

```

Output:



```

(dhanush@dhanush)-[~]
$ nano lab12a.c

(dhanush@dhanush)-[~]
$ gcc lab12a.c -o lab12a

(dhanush@dhanush)-[~]
$ ./lab12a
Enter Number of Pages: 2
Enter Reference String:
2
2
Enter Number of Frames: 2
Total Page Faults = 1

```

PROGRAM B: LRU PAGE REPLACEMENT

C PROGRAM

```
#include <stdio.h>

int main() {

int pages[50], frames[10], time[10];

int n, f, i, j;

int fault = 0, count = 0;

int found, pos, min;

printf("Enter Number of Pages: ");

scanf("%d", &n);

printf("Enter Reference String:
\n");


for(i = 0; i < n; i++)

scanf("%d", &pages[i]);

printf("Enter Number of Frames: ");

scanf("%d", &f);

for(i = 0; i < f; i++)

frames[i] =

-1;

for(i = 0; i < n; i++)

{

found = 0;

for(j = 0; j < f; j++)

{

if(frames[j] == pages[i])

{

count++;
```

```
time[j] = count;
found = 1;
break;
}
}
if(found == 0)
{
min = time[0];
pos = 0;
for(j = 0; j < f; j++)
{
```

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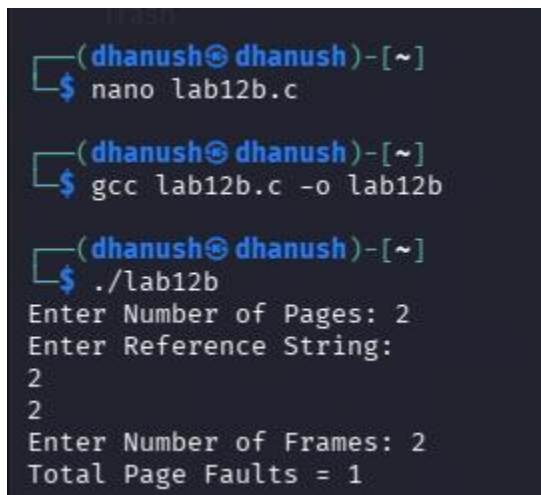
```
if(frames[j] == -1)
{
pos = j;
break;
}
if(time[j] < min)
{
min = time[j];
pos = j;
}
}
frames[pos] = pages[i];
```

```

count++;
time[pos] = count;
fault++;
}
}
printf("Total Page Faults = %d\n", fault);
return 0;
}

```

Output:



```

(dhanush@dhanush)-[~]
$ nano lab12b.c

(dhanush@dhanush)-[~]
$ gcc lab12b.c -o lab12b

(dhanush@dhanush)-[~]
$ ./lab12b
Enter Number of Pages: 2
Enter Reference String:
2
2
Enter Number of Frames: 2
Total Page Faults = 1

```

SHELL SCRIPT

```

#!/bin/bash

echo "LRU Page Replacement Demonstration"

pages=(7 0 1 2 0 3 0 4 2 3 0 3 2)

echo "Reference String: ${pages[@]}"

echo "LRU Algorithm Executed"

```

Output:

```
(dhanush@dhanush)-[~]
$ nano lab12b.sh

(dhanush@dhanush)-[~]
$ chmod +x lab12b.sh

(dhanush@dhanush)-[~]
$ bash lab12b.sh
LRU Page Replacement Demonstration
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
LRU Algorithm Executed
```

PROGRAM C: OPTIMAL PAGE REPLACEMENT

C PROGRAM

```
#include <stdio.h>

int main()
{
    int pages[50], frames[10];
    int n, f;
    int i, j, k, pos;
    int fault = 0;
    int found;

    printf("Enter Number of Pages: ");
    scanf("%d", &n);

    printf("Enter Reference String:
    \n");

    for(i = 0; i < n; i++)
        scanf("%d", &pages[i]);

    printf("Enter Number of Frames: ");
    scanf("%d", &f);

    for(i = 0; i < f; i++)
```

```
frames[i] =  
-1;  
for(i = 0; i < n; i++)  
{  
    found = 0;  
    for(j = 0; j < f; j++)  
    {  
        if(frames[j] == pages[i])  
        {  
            found = 1;  
            break;  
        }  
    }  
    if(found == 0)  
    {  
        for(j = 0; j < f; j++)  
        {  
            int future = 999;  
            for(k = i + 1; k < n; k++)  
            {  
                if(frames[j] == pages[k])  
  
                {  
  
                    future = k;  
                    break;  
                }  
            }  
        }  
    }  
}
```

```
}  
}  
if(j == 0 || future > pos)  
{  
    pos = future;  
}  
}  
  
frames[0] = pages[i];  
fault++;  
}  
}  
  
printf("Total Page Faults = %d\n", fault);  
  
return 0;  
}
```

Output:

```
(dhanush@dhanush)-[~]  
$ nano lab12c.c  
  
(dhanush@dhanush)-[~]  
$ gcc lab12c.c -o lab12c  
  
(dhanush@dhanush)-[~]  
$ ./lab12c  
Enter Number of Pages: 2  
Enter Reference String:  
1  
1  
Enter Number of Frames: 2  
Total Page Faults = 1
```


SHELL SCRIPT

```
#!/bin/bash
```

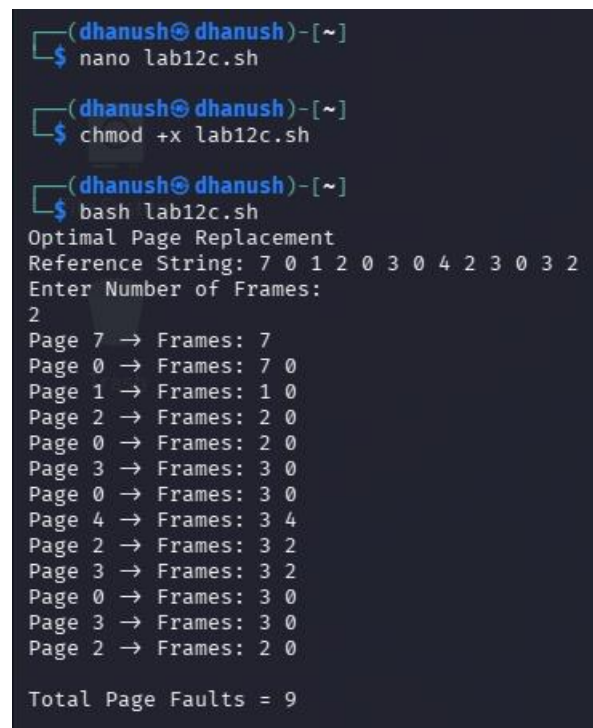
```
echo "Optimal Page Replacement Demonstration"
```

```
pages=(7 0 1 2 0 3 0 4 2 3 0 3 2)
```

```
echo "Reference String: ${pages[@]}"
```

```
echo "Optimal Algorithm Executed"
```

Output:



```
(dhanush@dhanush)-[~]
$ nano lab12c.sh

(dhanush@dhanush)-[~]
$ chmod +x lab12c.sh

(dhanush@dhanush)-[~]
$ bash lab12c.sh
Optimal Page Replacement
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
Enter Number of Frames:
2
Page 7 → Frames: 7
Page 0 → Frames: 7 0
Page 1 → Frames: 1 0
Page 2 → Frames: 2 0
Page 0 → Frames: 2 0
Page 3 → Frames: 3 0
Page 0 → Frames: 3 0
Page 4 → Frames: 3 4
Page 2 → Frames: 3 2
Page 3 → Frames: 3 2
Page 0 → Frames: 3 0
Page 3 → Frames: 3 0
Page 2 → Frames: 2 0

Total Page Faults = 9
```